18/12/23 M/S

USN. | | | | | | | | | | [18EEOE03]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Define Operations Research. Explain the various phases of Operations Research study. **07**

b. Solve by using graphical method the following linear programming problem. **07**

Max. $z=3x_1 - x_2$,

subject to constraints

$2x_1 + x_2 \geq 2$,

$x_1 + 3x_2 \leq 3$,

$x_2 \leq 4$, &

$x_1, x_2 \geq = 0$

2. a. Find the initial basic feasible solution of the following transportation problem using North West Corner Rule (NWCR). **06**

Warehouse / Factory	W_1	W_2	W_3	W_4	Factory Capacity
F_1	19	30	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Warehouse Requirement	5	8	7	14	34

b. A department head has 4 subordinates & 4 tasks have to be performed. Subordinates differ in efficiency & tasks differ in their intrinsic difficulty. Time each man would take to perform each task is given in the effectiveness matrix. How the tasks should be allocated to each person so as to minimize the total man-hours? **08**

		I	II	III	IV
TASKS	A	8	26	17	11
	B	13	28	4	26
	C	38	19	18	15
	D	19	26	24	10

3. a. Define the following terms with reference to a PERT chart: **06**

i. Total float

ii. Free float

iii. Independent float

b. A project has the following times schedule : **08**

Activity	Time in weeks	Activity	Time in weeks
1-2	4	5-7	8
1-3	1	6-8	1
2-4	1	7-8	2
3-4	1	8-9	1
3-5	6	8-10	8
4-9	5	9-10	7
5-6	4		

Construct PERT network and compute :

i. T_E & T_L for each event.

ii. Float for each activity.

iii. Critical path & its duration.

OR

4. a. Define the following terms with its notation. Represent the same in Time distribution curve : **06**

i. Optimistic time

ii. most likely time

iii. pessimistic time

b. A project has the following characteristics : **08**

Activity	Optimistic time(a)	Pessimistic time (b)	Most likely time (m)
1-2	1	5	1.5
2-3	1	3	2
2-4	1	5	3
3-5	3	5	4
4-5	2	4	3
4-6	3	7	5
5-7	4	6	5
6-7	6	8	7
7-8	2	6	4
7-9	5	8	6
8-10	1	3	2
9-10	3	7	5

Construct a PERT network. Find critical path & variance for each event.

5. a. Define the following terms **06**

i. Pay off matrix

ii. Two person zero sum game

iii. Saddle point

b. Solve the following (2 x 3) game graphically. **08**

			y_1	y_2	y_3
			I	**II**	**III**
	x	**I**	1	3	11
A	**$1-x_1$**	**II**	8	5	2

OR

6. **a.** The payoff matrix of a game is given below. Find the solution of the game to the player A & B. **07**

PLAYER A	**PLAYER B**				
	I	II	III	IV	V
I	-2	0	0	5	3
II	3	2	1	2	2
III	-4	-3	0	-2	6
IV	5	3	-4	2	-6

b. In a game of matching coins with two players, suppose one players wins Rs.2 when there are two heads & wins nothing when there are two tails; losses Re.1 when there are one head and one tail. Determine the payoff matrix, the best strategies for each player & the value of the game. **07**

7. **a.** Discuss the types of forecasting in decision making. **06**

b. A firm believes that its annual profit depends on its expenditures for research. The information for the preceding six years is given below. Estimate the profit when the expenditure is 6 units. **08**

Year	**Expenditure for Research (X)**	**Annual Profit (Y)**
1	2	20
2	3	25
3	5	34
4	4	30
5	11	40
6	5	31
7	6	?

OR

8. **a.** Briefly explain the steps in Delphi method. **07**

b. A firm uses simple exponential smoothing with $\alpha=0.2$ to forecast demand. The forecast for first week of January was 400 units, whereas actual demand turned out to be 450 units. **07**

i. Forecast the demand for second week of January.

ii. Assume that actual demand during second week of January turned out to be 460 units. Forecast the demand up to February third week, assuming the subsequent demands as 465,434,420, 498 & 462 units.